COMP8620

Advanced Topics in AI: Intelligent Robotics

Assignment 2 Report

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Generative AI Disclosure

OpenAI Codex was used to prepare the report template, review and revise the Part A answers, implement the first Part B solver and tests, design and run the first experiments, and draft report text, pseudocode and figures. Anthropic Claude Code (Claude Opus 5.5) was then used to review the Part B code and experiments, implement the revised solver (risk-aware guidance and admissible exploration), redesign and rerun the Part B experiments, analyse the measured results, and revise the Part B text, tables and figures. The supplied simulator's transition and reward methods were preserved; their use and the headless import changes are documented in Part B. Original-language prompts from all these interactions, including supplied draft answers, are recorded verbatim in `AI_prompts.txt` in the submitted source archive.

Part A: Theory (30 points)

Question 1: Two-state MDP abstraction (10 points)

No.

Let $m = \lceil k/2 \rceil$, and define

$$\phi(s) = \begin{cases} L, & 1 \le s \le m, \\ R, & m+1 \le s \le k. \end{cases}$$

An exact MDP abstraction requires a transition function depending only on the abstract state and action while preserving the original motion probabilities. In particular, for every action a, transition consistency requires

$$\phi(s_1) = \phi(s_2) \implies \sum_{u:\, \phi(u)=R} T(s_1, a, u) = \sum_{u:\, \phi(u)=R} T(s_2, a, u).$$

For $k > 4$, consider $s_1 = m - 1$ and $s_2 = m$, both in L. Under $a = \text{moveRight}$,

$$m-1 \xrightarrow{0.7} m \in L, \qquad m-1 \xrightarrow{0.3} m-2 \in L,$$
$$m \xrightarrow{0.7} m+1 \in R, \qquad m \xrightarrow{0.3} m-1 \in L.$$

Hence,

$$P(R \mid s_1, a) = \sum_{u=m+1}^{k} T(m-1, a, u) = 0,$$
$$P(R \mid s_2, a) = \sum_{u=m+1}^{k} T(m, a, u) = T(m, a, m+1) = 0.7.$$

Since $0 \neq 0.7$, transition consistency fails: no single $T'(L, \text{moveRight}, R)$ can represent both states exactly.

Therefore, the proposed abstraction cannot preserve the original motion uncertainty and is not an exact MDP abstraction.

Question 2: Alpha-vector dominance (10 points)

Let $p = b(s_1)$. Since $b(s_1) + b(s_2) = 1$,

$$b = (p, 1-p), \qquad 0 \le p \le 1.$$

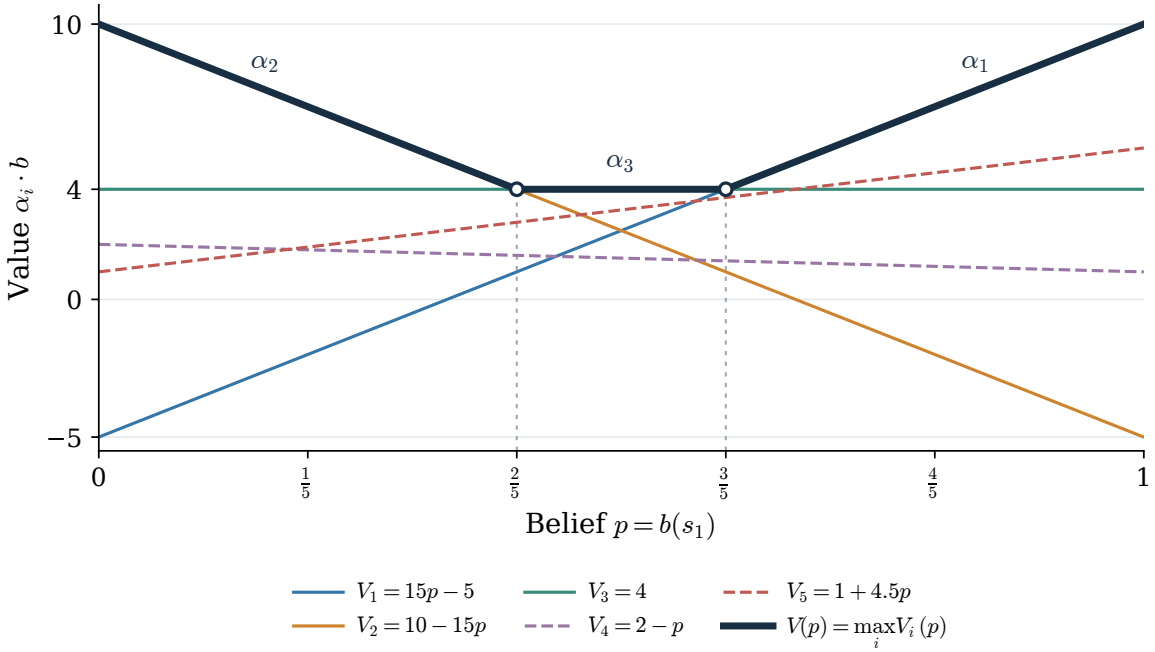
For each $\alpha_i \in \Gamma$, let $V_i(p) = \alpha_i \cdot b$. Then

$$\begin{aligned} V_1(p) &= 10p - 5(1-p) = 15p - 5, \\ V_2(p) &= -5p + 10(1-p) = 10 - 15p, \\ V_3(p) &= 4p + 4(1-p) = 4, \\ V_4(p) &= p + 2(1-p) = 2 - p, \\ V_5(p) &= 5.5p + (1-p) = 1 + 4.5p. \end{aligned}$$

(a) Maximal belief intervals (4 points)

The upper envelope is

$$V(p) = \max_{\alpha_i \in \Gamma} V_i(p).$$



The relevant intersections are

$$\begin{aligned} V_2(p) = V_3(p) &\iff 10 - 15p = 4 \iff p = \frac{2}{5}, \\ V_1(p) = V_3(p) &\iff 15p - 5 = 4 \iff p = \frac{3}{5}. \end{aligned}$$

The remaining vectors are never maximal, since

$$V_4(p) = 2 - p < 4 = V_3(p), \qquad 0 \le p \le 1,$$

and, using $V_1(p) - V_5(p) = 10.5p - 6$,

$$\begin{cases} V_5(p) \le 1 + 4.5 \times \frac{3}{5} = 3.7 < 4 = V_3(p), & 0 \le p \le \frac{3}{5}, \\ V_1(p) - V_5(p) \ge 10.5 \times \frac{3}{5} - 6 = 0.3 > 0, & \frac{3}{5} \le p \le 1. \end{cases}$$

Since V_2 decreases, V_1 increases, and $V_3 = 4$,

$$V(p) = \begin{cases} 10 - 15p, & 0 \le p < \frac{2}{5}, \\ 4, & \frac{2}{5} \le p \le \frac{3}{5}, \\ 15p - 5, & \frac{3}{5} < p \le 1. \end{cases}$$

Therefore, the maximal belief intervals are

$$\boxed{\begin{aligned} \alpha_1 &: \left[\frac{3}{5}, 1\right], \\ \alpha_2 &: \left[0, \frac{2}{5}\right], \\ \alpha_3 &: \left[\frac{2}{5}, \frac{3}{5}\right], \\ \alpha_4, \alpha_5 &: \varnothing. \end{aligned}}$$

At $p = \frac{2}{5}$, α_2 and α_3 tie; at $p = \frac{3}{5}$, α_1 and α_3 tie.

(b) Pointwise dominance (6 points)

An α-vector can be removed if, at every belief, another vector has strictly greater value:

$$\forall b \in B, \quad \exists \alpha' \in \Gamma : \alpha \cdot b < \alpha' \cdot b.$$

For α_4,

$$V_4(p) = 2 - p < 4 = V_3(p), \qquad \forall p \in [0, 1].$$

Thus, α_4 is strictly dominated by α_3.

For α_5, the dominating vector may depend on p. From part (a),

$$\begin{cases} V_5(p) < V_3(p), & 0 \le p \le \frac{3}{5}, \\ V_5(p) < V_1(p), & \frac{3}{5} < p \le 1. \end{cases}$$

These intervals cover $[0, 1]$, so α_5 also satisfies the removal condition.

The remaining vectors α_2, α_3, and α_1 are uniquely maximal at $p = 0$, $p = \frac{1}{2}$, and $p = 1$, respectively. Hence, none satisfies the removal condition.

Therefore,

$$\boxed{\Gamma_{\text{remove}} = \{\alpha_4, \alpha_5\}, \qquad \Gamma_{\text{keep}} = \{\alpha_1, \alpha_2, \alpha_3\}.}$$

Question 3: MDP versus POMDP for pedestrian avoidance (10 points)

Let c_t, v_t be the car's position and speed, and p_t^i the position of pedestrian i. Define

$$x_t = (c_t, v_t, p_t^1, \ldots, p_t^n).$$

Positions are observed accurately, and the car measures its own speed. It chooses an action a_t (accelerate, maintain speed, or brake) to reach G while avoiding collisions.

(a) Scenario suitable for an MDP

Suppose pedestrians follow known stochastic movement rules: the probabilities of crossing or continuing along the sidewalk depend only on the current observable configuration, with no persistent hidden intention. These memoryless dynamics give

$$P(x_{t+1} \mid x_{0:t}, a_{0:t}) = P(x_{t+1} \mid x_t, a_t).$$

Thus, x_t is Markov and fully observed, so an MDP with state $s_t = x_t$ is suitable despite the stochastic pedestrian motion.

(b) Scenario suitable for a POMDP

Now suppose each pedestrian has a persistent but unobservable destination g_i, such as crossing the road or continuing along the sidewalk. Let $\mathbf{g} = (g_1, \ldots, g_n)$, and assume motion is Markov conditional on these destinations. The complete state and observation are

$$s_t = (x_t, \mathbf{g}), \qquad o_t = x_t.$$

At the same observed configuration, two possible joint destinations can yield different motion distributions:

$$P(x_{t+1} \mid x_t, a_t, \mathbf{g}^{(1)}) \neq P(x_{t+1} \mid x_t, a_t, \mathbf{g}^{(2)}).$$

Different motion histories can produce the same x_t but different beliefs about $\mathbf{g}$. Hence, x_t alone need not be Markov. The car maintains

$$b_t(\mathbf{g}) = P(\mathbf{g} \mid x_{0:t}, a_{0:t-1})$$

and predicts motion using

$$P(x_{t+1} \mid x_t, b_t, a_t) = \sum_{\mathbf{g}} P(x_{t+1} \mid x_t, a_t, \mathbf{g})\, b_t(\mathbf{g}).$$

The belief is updated as new movements are observed. A POMDP is suitable because it uses both the observed configuration and the belief over hidden destinations to select driving actions.

Part B: Multi-drone planning under uncertainty (35 points)

Question 1: MDP formulation (3 points)

Let $\mathcal{G} = \{0, \dots, n_x - 1\} \times \{0, \dots, n_y - 1\} \times \{0, \dots, n_z - 1\}$, with obstacles O and known drone goals g_i. The finite-horizon MDP is $(\mathcal{S}, \mathcal{A}, T, R, \gamma)$.

States and actions. A live state is

$$s = ((p_i, r_i)_{i=1}^{N}, t), \qquad 0 \le t < H,$$

where p_i is drone i's cell and r_i indicates that it has reached its goal. Reached drones remain at their goals and occupy those cells. The state space also contains an absorbing state $\perp$, entered after a collision, all goals being reached, or the step limit H. Each active drone commands one of $B = 26$ neighbour offsets (or $B = 8$ with fixed altitude), giving $\mathcal{A} = \{0, \dots, B-1\}^N$. There is no hover action; reached drones ignore their commands.

Transitions. Let $q_i(y_i \mid p_i, r_i, a_i)$ be the supplied, normalised single-drone target law (Appendix F). For a reached drone, $q_i(y_i \mid p_i, 1, a_i) = \mathbf{1}\{y_i = p_i\}$. Raw targets $y = (y_1, \dots, y_N)$ are conditionally independent. Let $F(s, y)$ apply the simulator's position and flag updates and return $\perp$ on termination. Then

$$T(s, a, s') = \sum_{y : F(s,y) = s'} \prod_{i=1}^{N} q_i(y_i \mid p_i, r_i, a_i), \qquad T(\perp, a, \perp) = 1.$$

An obstacle hit or shared returned cell terminates the mission; cell swaps and diagonal crossings are permitted by the supplied collision rule.

Reward and objective. For a sampled outcome y, let K_O count obstacle hits, K_V count drones in shared-cell groups, and Δr_i indicate a newly reached goal. The simulator returns

$$\widetilde{r}(s, a, y) = N c_{\text{step}} + C(K_O + K_V) + G \sum_i \Delta r_i, \qquad c_{\text{step}} < 0,\ C < 0,\ G > 0.$$

Thus $R(s, a) = \mathbb{E}_y[\widetilde{r}(s, a, y) \mid s, a]$ and $R(\perp, a) = 0$. The terminal transition receives its reward before absorption. The objective is

$$\max_{\mu}\ \mathbb{E}_{\mu}\left[\sum_{t=0}^{\tau-1} \gamma^t \widetilde{r}_t\right], \qquad \tau = \min\{T_{\text{success}}, T_{\text{collision}}, H\}.$$

This encourages fast completion and penalises collisions.

Question 2: Approximate online solver (4 points)

I use UCT/MCTS with progressive widening [1, 4, 5], re-planned after every executed step. The supplied generative model samples outcomes without enumerating the state space or all $26^4 = 456{,}976$ joint actions.

Guidance. Cached Dijkstra maps J_i trade path length against one-step obstacle-slip risk. For a command from p_i towards q, the score is

$$J_i(q) + 0.025\|q - g_i\|_1 + \frac{0.3}{\max\{1, d(q)\}} + \lambda \ell(p_i) \sum_{\substack{k \neq i \\ r_k = 0}} \sigma(p_k) \mathbf{1}\{q \in \mathcal{N}(p_k)\}.$$

Here d is the supplied obstacle-distance transform; ℓ and σ are the normalised intended and per-neighbour slip probabilities, and λ converts collision loss into step-cost units (Appendix F).

The last term is an inter-drone risk surrogate. $\mathcal{N}$ contains physical neighbours at the allowed altitude. Out-of-grid and obstacle targets receive prohibitive costs; frozen or already reserved cells receive 10^9. GREEDY processes drones in index order and minimises this score; SAMPLE uses random order and samples among the four cheapest commands within 1.05 of the minimum, with weights $e^{-3(\text{score}-\min)}$. EXPLORE samples uniformly from free, in-grid, unreserved targets when available. Figure 1 shows these maps on two single-drone cases. Slip risk ρ is confined to cells beside obstacles, so the risk-weighted cost-to-go routes the greedy policy around a cluttered block that the step distance would cut through (top). Behind a wall every route must pay for the one gap (bottom); there the executed run hesitates before crossing, the horizon effect noted in the limitations.

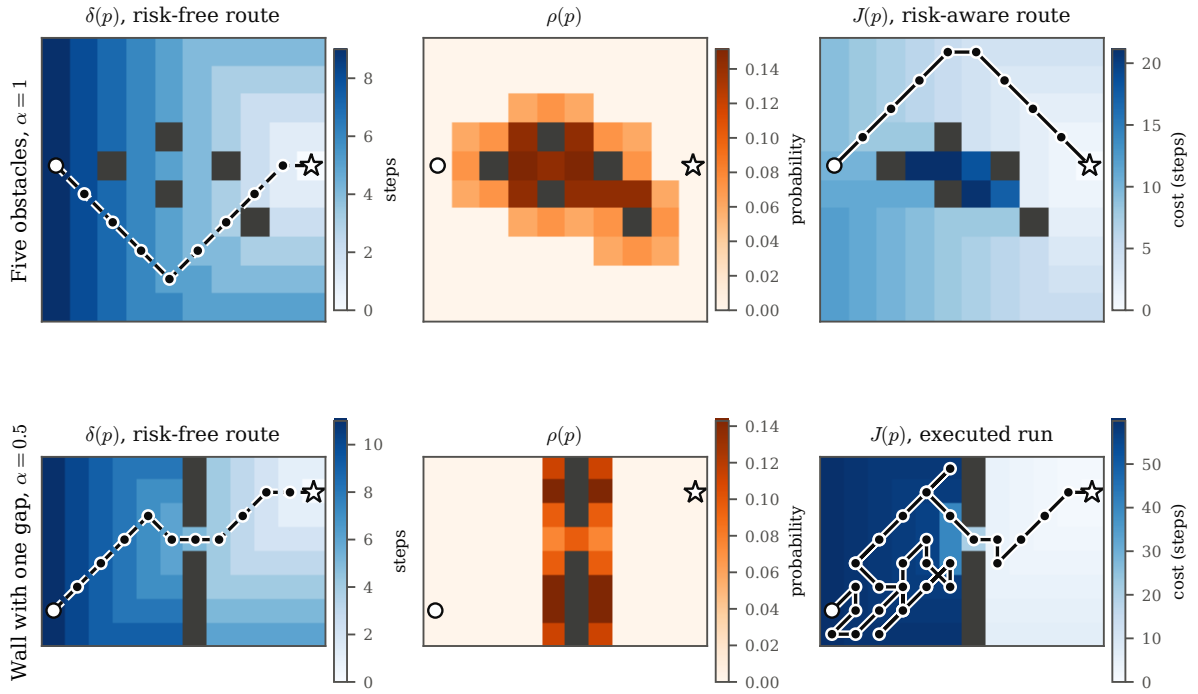


Figure 1: Guidance maps for two fixed-altitude single-drone cases (Appendix E). Left: breadth-first steps $\delta(p)$ with the greedy route of risk-free guidance ($w = 0$). Middle: departure risk $\rho(p)$, the probability that the next move slips into an obstacle. Right: risk-weighted cost-to-go $J(p)$ with the risk-aware greedy route (top) or the executed seed-0 run (bottom). Dark cells are obstacles; circle: start; star: goal. Greedy routes assume no slips; colour scales differ by panel.

The four MCTS components.

(1) **Selection.** Try unvisited admitted actions first; otherwise maximise

$$Q(s,a) + c_{\text{UCT}}\sqrt{\frac{\log(N(s)+1)}{N(s,a)}}, \qquad c_{\text{UCT}} = 0.35.$$

Returns are normalised by $\kappa = \max\{1, N(G + 2|C|)\}$.

(2) **Expansion.** The action limit is $\min\{\lceil 0.5\sqrt{N(s)+1}\rceil, \prod_{i:r_i=0} |U_i^{\text{adm}}|\}$. Admit GREEDY first, then EXPLORE with probability 0.15 and SAMPLE otherwise (at most 12 attempts for a new action). Each distinct sampled successor has its own child.

(3) **Simulation.** Roll out for at most eight steps, choosing EXPLORE with probability 0.1,

otherwise GREEDY. Terminal outcomes have zero continuation value; other cutoffs add

$$\widehat{V}(s,h) = Nc_{\text{step}} \sum_{j=0}^{D-1} \gamma^j + G \sum_{\substack{i:r_i=0 \\ 1\le\delta_i\le h}} \gamma^{\delta_i - 1}, \qquad D = \min\{h, \max_{i:r_i=0} \delta_i\}.$$

δ_i is the physical breadth-first distance, set to ∞ when unreachable or when frozen goal occupants disconnect the route. The latter connectivity check is cached on maps of at most 20,000 cells. Tree depth is capped at eight.

(4) **Backpropagation.** Back up $v = R_{\text{sample}}/\kappa + \gamma v'$; increment edge and node visits and update the edge mean and sample variance using Welford's recurrence.

Root decision. Let a_g be the greedy action and $\hat{\sigma}^2$ the pooled within-action variance. Among visited actions, define

$$m(a) = Q(a) - Q(a_g) - z_{\text{sw}}\hat{\sigma}\sqrt{\frac{1}{N(a)} + \frac{1}{N(a_g)}}, \qquad z_{\text{sw}} = 1.$$

Execute the action with largest positive $m(a)$; otherwise keep a_g. With no completed trial, use a_g directly. This variance-aware switching heuristic has no formal policy-improvement guarantee. Parameters were selected on separate development layouts (Appendix B).

Listing 1: Online UCT; rollout and admission rules are defined above.

```
PROCEDURE PLAN(s, budget, K = 2000)
    deadline <- NOW + max(0, 0.97 * min(budget, 10) - 0.002)
    a_g <- GREEDY(s); root <- empty node; h <- H - t; sims <- 0
    WHILE NOW < deadline AND sims < K AND h > 0 AND not all reached:
        TRY: SEARCH(root, s, h, 0); sims += 1
        ON Deadline: BREAK
    RETURN root decision above, or a_g if no trial completed

PROCEDURE SEARCH(node, s, h, depth)
    IF h = 0 OR all reached: RETURN 0
    IF depth >= 8: RETURN ROLLOUT(s, h)
    admit an action under the expansion rule, if permitted
    a <- unvisited action first, otherwise argmax UCB
    (s', r, done) <- SIMULATE(s, a) // check deadline before call
    IF done: v <- 0
    ELSE IF child(node, a, s') is new:
        create child; v <- ROLLOUT(s', h - 1)
    ELSE: v <- SEARCH(child(node, a, s'), s', h - 1, depth + 1)
    v <- r / kappa + gamma * v
    update visits, mean and variance; RETURN v
```

ROLLOUT discounts sampled rewards and the terminal/cutoff value defined above. It may stop at the deadline and return a leaf estimate; a deadline exception before a simulator call aborts that trial. The planner tracks the remaining horizon because `simulate` does not enforce the execution step limit.

Question 3: Implementation and demonstration (8 points)

`run_u8076655.py` executes the closed-loop planner. The supplied simulator is vendored at revision `ea3cbfe`; AST checks verify that its core dynamics and rewards are unchanged [6]. The planner uses `get_config()`, `simulate()` and `num_actions`, with separate planning and execution RNGs. From the extracted archive's top-level directory:

```
python -m pip install -r requirements.txt
python run_u8076655.py --config <scenario.yaml>
```

Each step prints progress; `-visualize` enables the optional Vedo viewer. Planning is serial, with a default 9.5 s allowance and 2,000-trial cap. The clock is checked before every simulator call, with a margin for returning the action. Local validation results follow. All 34 unit tests pass. A separate exhaustive check of the free $2 \times 2 \times 1$ two-drone model covers 11 admissible nonterminal joint states, 704 state–action pairs and all 6,048 positive-probability outcomes. Single-process runs with the default 9.5-second allowance (seed 0) give:

Task	Status	Steps	Sims/step	Mean (s)	Max (s)
Open 2D, 1 drone, 8 actions	success	11	2,000	5.37	9.14
Supplied example, 2 drones	success	17	2,000	5.43	7.39
Strong wind, 1 drone, $\alpha = 3$	success	18	2,000	4.69	6.29
Four 3D, 5 obstacles	success	13	1,983	6.34	9.21

A forced 10-second call with the trial cap raised to 100,000 returns in 9.699 seconds. On slower hardware the wall-clock deadline, not the trial cap, ends the search; these are measurements on one machine, not a hard real-time guarantee.

Question 4: Experimental investigation (20 points)

Experimental design

The main evaluation uses 2,000 simulations per decision; S3 includes ablations and a zero-search policy, and S4 varies the simulation cap. Experimental runs use a relaxed wall-clock cap to isolate search effort. The demo instead stops at the first of its time and trial limits, so these are different budget protocols. Conditions share execution seeds 0–19 unless noted, with comparisons paired within each scenario. Development layouts were excluded from evaluation. Original run hashes are retained in the metadata; subsequent planner patches are checked separately before these results are reused: replaying all 137 reported seed-0 episodes with the submitted planner reproduced 137 of them step for step (`results/planner_equivalence.json`).

Set	Controlled scenarios	Question and expectation
S1 (540)	3^3 factorial: $12 \times n_y \times 3$ grids, $n_y = 5, 10, 20$; 6/18/26 commands per drone ($\|A\| = 36/324/676$); 0/2/5 nested obstacles. Two drones, $\alpha = 0.25$, $H = 60$.	Rank the three factors and test interactions. Off-route cells should have little effect; richer control may help; clutter adds detours and slip risk.
S2 (160)	S1 clutter layouts at $\alpha = 0.01$ with 6/26 commands; duplicate command labels $\times 2, \times 4$ without changing physics.	Separate geometric detours from noise, and control authority from branching.
S3 (880)	Five ablations and the search-free policy at all eight low/high S1 corners. Uniform proposals use ten seeds; others use twenty.	Measure how component effects change with each complexity factor.
S4 (400)	16/64/256/1,024/2,000 simulations on open 1-drone, strong-wind 1-drone, cluttered 2-drone and 3D 4-drone tasks.	Test whether additional search improves completion as tasks get harder.
S5 (160)	$N = 1, 2, 3, 4$ crossed with $\alpha = 0.05, 0.25, 1, 2$; one random $8 \times 6 \times 3$ layout per pair, ten seeds each.	Check performance beyond the factorial and joint growth of state/action spaces.

Table 1: Evaluation sets; parentheses give episode counts. A further 360 regression episodes are listed in Appendix D.

A further 520 search-free runs pair the base policy with the completed regression and generated suites (Appendix D). Seven supplementary end-to-end cases (280 runs, each paired with the base policy) cover solvable fixed-altitude single-drone maps, fixed-altitude drones on different layers, and a goal inside a teammate's only corridor (Appendix E).

Metrics. With F free cells, there are up to F^N positional configurations and B^N joint commands. Success means all goals reached within H without collision. Penalised completion length is $L =$ steps on success and $L = H$ on failure. Success proportions have Wilson 95% intervals; continuous means have Student-t intervals $\bar{x} \pm t_{0.975,n-1} s/\sqrt{n}$. High-minus-low factor effects average the other factors within each seed; component effects pair each ablation with the full solver before averaging corners. Their 95% intervals use seed blocks. Full condition means, returns and search diagnostics are in Appendix G–J. Episodes ran on macOS-26.6.2 (`x86_64`), Python 3.12.9, NumPy 1.26.4, SciPy 1.15.1, PyYAML 6.0.2. Each episode plans in one process and one thread; independent episodes ran in parallel worker processes. Because search effort is fixed by the simulation cap, machine load changes wall time but not the decisions.

S1: effects of the three complexity factors

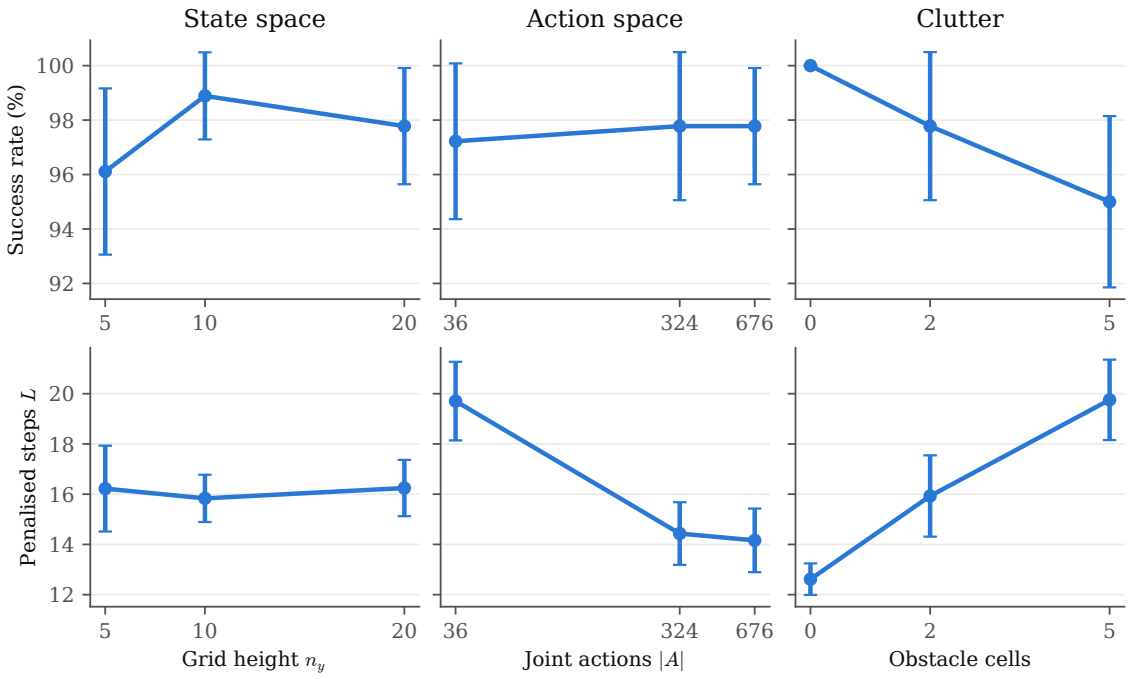


Figure 2: Factorial marginal means; bars are 95% intervals over 20 seed blocks.

Factor (high minus low)	ΔL (95% CI)	Δsuccess (pp; 95% CI)
State space (n_y 5→20)	+0.0 [-1.4, +1.4]	+1.7 [-1.4, +4.7]
Action space ($\|A\|$ 36→676)	-5.5 [-7.0, -4.1]	+0.6 [-2.6, +3.7]
Clutter ($\|O\|$ 0→5)	+7.1 [+5.7, +8.6]	-5.0 [-8.1, -1.9]

Clutter causes the largest degradation within these tested levels. Adding five obstacles raises L by $+7.1$ steps $[+5.7, +8.6]$ and changes success by -5.0 percentage points $[-8.1, -1.9]$. The solver succeeds in 97.6% of S1 episodes; all 13 failures are collisions and none reaches the horizon.

State-space padding has little observed effect. For $n_y = 10$ versus 20, 120 of 180 paired episodes are identical. The extra off-route cells do not affect these runs; this tests padding, not the effect of longer routes or additional drones. Mean map setup grows from 52 to 230 ms ($n_y = 5$ versus 20), outside per-decision planning.

Richer commands shorten routes. Changing 6 to 26 directions gives $\Delta L = -5.5$ $[-7.0, -4.1]$; the success difference is not statistically resolved. The action–clutter interaction is -7.7 $[-11.0, -4.4]$ steps, consistent with diagonal/vertical moves easing obstacle avoidance. Neither state-related interaction is resolved ($S \times C$: $[-3.8, +4.7]$; $S \times A$: $[-3.1, +4.3]$).

S2: geometry, noise and action branching

At $n_y = 10$, the clutter effect averaged over the two command sets is $+9.4$ steps $[+6.4, +12.5]$ at $\alpha = 0.25$, versus $+1.5$ $[+1.2, +1.8]$ at $\alpha = 0.01$, where all episodes succeed. Thus geometry still adds detour cost, while noise substantially increases the observed clutter penalty. The supplied transition law raises slip probability near obstacles; its no-obstacle distance-transform convention is noted in Appendix F.

Duplicating command labels up to $|A| = 10{,}816$ changes L by only $+0.1$ $[-0.1, +0.2]$ and leaves the admitted root-action count near 23. This separates pure label branching from the extra control authority in S1. Detailed controls, including the visit share of the *executed* action, appear in Appendix H.

S3: component effects as complexity increases

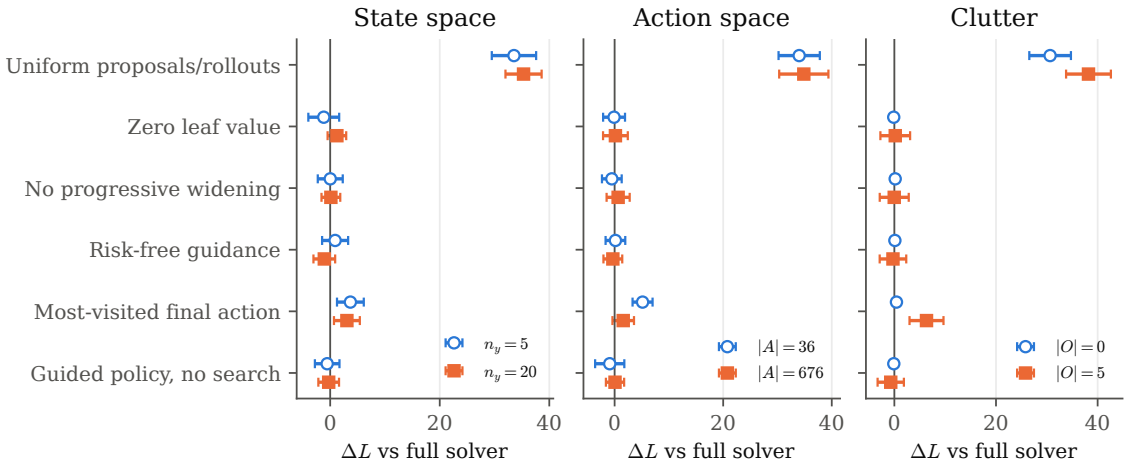


Figure 3: Paired change in L after removing a component at low/high factor levels (open/filled markers); 95% seed-block intervals. Positive values favour the full solver.

Guidance in expansion and rollout has the largest measured effect. Uniform proposals/rollouts raise L by $+34.4$ $[+31.5, +37.4]$ and change success by -50.0 pp. The loss grows from $+30.6$ without obstacles to $+38.2$ with five; the clutter interaction excludes zero ($[+1.4, +13.7]$), whereas its state/action interactions do not. This ablation removes both route guidance and admissible-target filtering.

Root choice also matters: most-visited selection raises L by $+3.4$ $[+1.7, +5.0]$. The increase is larger with clutter ($[+2.5, +9.3]$ interaction interval) and smaller with richer commands ($[-5.5, -1.6]$). These comparisons favour the chosen rule on these tasks.

Widening, leaf estimates and risk surcharges have no statistically resolved effects across these corners (full estimates in Appendix C). Routes are roughly eleven steps, comparable to the sampled tree depth plus eight-step rollout, which may limit the leaf estimate's contribution. The risk-free ablation retains route guidance; its result therefore does not contradict the large loss from removing all guidance. Risk-aware tuning gains on development layouts (Appendix B) are exploratory evidence.

Search versus its base policy. Removing search changes L by -0.4 $[-1.7, +0.9]$, with no resolved interaction. The evaluated completion performance is largely explained by the guided base policy. In the additional regression/generated comparison, multi-drone search rescues 12 episodes and loses 4; the pooled difference is inconclusive (Appendix D).

S4: search budget

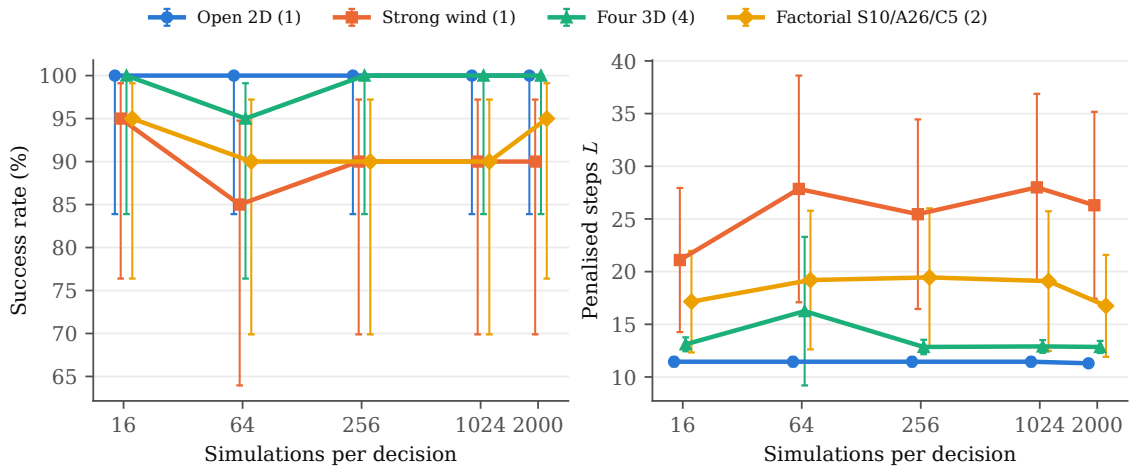


Figure 4: Success and L versus simulations per decision; 20 runs per point, with Wilson and t 95% intervals. Drone counts are in parentheses.

There is no consistent improvement from 16 to 2,000 simulations across these four tasks. In the four-drone case, success is 20/20 and 20/20 respectively, while mean new-node depth grows from 1.9 to 2.7 and the fraction keeping the base action falls from 0.81 to 0.61. Search therefore changes decisions without a demonstrated completion benefit here. Overlapping intervals do not establish budget equivalence.

S5: generated layouts

Across the crossed drone-count/noise set, 148/160 episodes succeed. At $\alpha \leq 0.25$, all 80/80 runs succeed, including 20/20 with four drones; four drones at $\alpha = 2$ succeed in 3/10. This indicates sensitivity to noise in the tested layouts, while increasing N jointly enlarges state and action spaces. Mean new-node depth is 3.1 for one drone and 2.4 for four, consistent with less successor reuse when joint outcomes multiply. Each (N, α) pair has only one layout, so the marginals in Appendix J do not isolate layout effects.

Conclusions and limitations

Within the factorial's tested levels, clutter causes the largest degradation; extra control directions help, while off-route state padding has little observed effect. Guided expansion/rollout and the root decision rule contribute most, particularly in clutter. Additional search has no demonstrated completion benefit over the guided base policy on these tasks.

The ranking is specific to two drones, $\alpha = 0.25$ and the chosen layouts. The t intervals for bimodal L are approximate, and generated-layout replication is limited. Fixed-simulation experiments do not reproduce a binding demo deadline on every machine. Static guidance does not model detours around frozen drones beyond the leaf disconnection check; observed collisions have not been proved unavoidable. The leaf estimate charges no collision risk, so search can postpone an unavoidable risky passage beyond its horizon: in the wall-gap case (Figure 1, bottom right) successful runs averaged 39 steps for an 11-step route. A risk-charged leaf shortened those runs in a small probe (26 to 19 steps) without raising success (65/80 versus 69/80 over eight tasks), so it was not adopted. Stronger coordination tasks and outcome widening or open-loop trees would be useful follow-up tests.

A Prompt log and source inventory

The archive's top-level folder is `u8076655_A2_code`; the file `AI_prompts.txt` records the original-language prompts.

File/folder	Purpose
`run_u8076655.py`	Demo entry point, closed-loop execution and progress output.
`planner.py`	Risk-aware budgeted UCT, guidance maps, leaf values and diagnostics.
`environment.py`	Headless hooks, input checks and experimental action masks.
`multi_drone*.py`	Supplied simulator and configuration class (unchanged core).
`experiments.py`	Simulation-capped, resumable experiment runner.
`tests/`	Unit tests for model semantics, guidance and planning behaviour.
`environments/`	All 92 scenario entries (`dev/`: development set) and the manifest.
`results/`	Raw data, provenance, verification and seed-0 traces (`dev/`: tuning).
`scripts/`	Scenario generation, experiments, analysis, validation and packaging.
`report/`	LaTeX sources, generated tables and figures.

B Development-set parameter selection

Sixteen development layouts (`environments/dev/`: $10 \times 10 \times 5$ or $10 \times 10 \times 1$ grids, 1–4 drones crossed with $\alpha \in \{0.25, 0.5, 1, 2\}$, 0–5 random obstacles, generator seed 4620) were used with ten seeds and 500 simulations per decision, before any reported experiment; none of them appears in Question 4. Round 1 varied the risk weight w and c_{UCT} under the then most-visited choice and widening 1.5; round 2 varied the anchored choice. Differences are paired against the adopted setting (`anchor_z1_w0.5`), averaged over layouts within each seed ($n = 10$ blocks). Selecting the best of twelve settings on these layouts is optimistic for them, which is why all reported results use other scenarios.

Table 2: Development-set results ($n = 160$ episodes per setting). Obst./Veh.: episodes ending in obstacle / inter-drone collisions.

Rd.	Setting	Success	Obst.	Veh.	L	Δsuccess (pp)	ΔL
1	risk0	125/160	20	15	24.3 ± 3.1	-9.4 [-15.4, -3.3]	+4.5 [+2.1, +7.0]
1	risk0.3	130/160	19	8	23.4 ± 2.9	-6.2 [-12.2, -0.3]	+3.7 [+1.2, +6.1]
1	risk1	129/160	20	9	23.4 ± 3.0	-6.9 [-13.4, -0.4]	+3.7 [+0.6, +6.8]
1	risk3	126/160	23	7	23.8 ± 3.0	-8.8 [-15.5, -2.0]	+4.1 [+1.2, +7.0]
1	risk1_c0.1	129/160	21	8	23.0 ± 2.9	-6.9 [-12.2, -1.5]	+3.3 [+1.4, +5.2]
1	risk1_c1	123/160	28	8	25.1 ± 3.2	-10.6 [-15.4, -5.9]	+5.4 [+3.3, +7.5]
1	greedy_risk0	121/160	17	22	23.9 ± 3.2	-11.9 [-15.8, -8.0]	+4.2 [+2.0, +6.4]
1	greedy_risk1	138/160	19	4	19.8 ± 2.6	-1.2 [-5.4, +2.9]	+0.1 [-1.5, +1.6]
2	anchor_z1	133/160	19	8	21.3 ± 2.8	-4.4 [-8.6, -0.1]	+1.6 [-0.1, +3.3]
2	anchor_z2	136/160	18	6	20.7 ± 2.7	-2.5 [-7.3, +2.3]	+0.9 [-0.6, +2.4]
2	anchor_z1_eps0	133/160	21	6	21.4 ± 2.8	-4.4 [-7.4, -1.4]	+1.7 [+0.6, +2.8]
2	anchor_z1_w0.5	140/160	13	9	19.7 ± 2.5	—	—

C Component results by factorial corner

Table 3: Paired component effects at the eight S1 corners. Entries give mean changes with 95% intervals; factor columns show low/high-level means. * marks an interaction whose interval excludes zero.

Variant	ΔL	Δsuccess (pp)	n_y 5 / 20	$\lvert A\rvert$ 36 / 676	$\lvert O\rvert$ 0 / 5
Uniform proposals/rollouts	+34.4 [+31.5, +37.4]	-50.0 [-56.0, -44.0]	+33.5 / +35.3	+34.0 / +34.9	+30.6 / +38.2*
Zero leaf value	+0.0 [-1.5, +1.5]	+0.6 [-2.4, +3.6]	-1.2 / +1.2	-0.1 / +0.1	-0.1 / +0.2
No progressive widening	+0.1 [-1.4, +1.5]	+0.0 [-3.3, +3.3]	+0.0 / +0.1	-0.5 / +0.7	+0.2 / -0.0
Risk-free guidance	-0.1 [-1.4, +1.2]	-1.9 [-4.7, +1.0]	+0.9 / -1.1	+0.1 / -0.3	+0.1 / -0.3
Most-visited final action	+3.4 [+1.7, +5.0]	-1.9 [-6.2, +2.5]	+3.7 / +3.0	+5.2 / +1.6*	+0.4 / +6.3*
Guided policy, no search	-0.4 [-1.7, +0.9]	-0.6 [-3.6, +2.4]	-0.6 / -0.3	-0.9 / +0.1	-0.1 / -0.7

Table 4: Successes per corner (full solver: factorial runs, $n = 20$; uniform proposals: $n = 10$; other variants: $n = 20$). Corner labels give n_y, intended directions per drone and obstacle count.

Variant	5/6/0	5/6/5	5/26/0	5/26/5	20/6/0	20/6/5	20/26/0	20/26/5
Full solver	20/20	18/20	20/20	19/20	20/20	20/20	20/20	19/20
Uniform proposals/rollouts	9/10	0/10	9/10	0/10	10/10	0/10	9/10	0/10
Zero leaf value	20/20	20/20	20/20	18/20	20/20	20/20	20/20	19/20
No progressive widening	20/20	19/20	20/20	18/20	20/20	20/20	20/20	19/20
Risk-free guidance	20/20	16/20	20/20	19/20	20/20	19/20	20/20	19/20
Most-visited final action	20/20	19/20	20/20	18/20	20/20	19/20	20/20	17/20
Guided policy, no search	20/20	18/20	20/20	19/20	20/20	19/20	20/20	19/20

D Regression coverage

The 18 named scenarios each use 20 seeds and a fixed cap of 2,000 simulations per decision. "No search" is the search-free base policy on the same seeds (baseline suite). Infeasible cases and the one-step horizon case are expected failures; stochastic collisions in feasible cases remain recorded failures.

Across feasible episodes with at least one executed step in the regression and generated suites, there are 240 multi-drone pairs: the full solver succeeds in 226 and the base policy in 218. Search rescues 12 episodes and loses 4. Removing search changes L by $+1.3$ $[-0.4, +3.0]$. The 180 single-drone pairs have 157 successes under each policy ($\Delta L = -0.4$ $[-1.6, +0.7]$). These exploratory intervals pool paired episode differences across layouts; they are not the seed-block intervals used for S1–S3 and do not establish an advantage for search.

Table 5: Named regression scenarios. L shows mean ±95% CI half-width; its failure penalty is the scenario's own step limit.

Scenario	Success	No search	Obst.	Veh.	L	Calls/step
`demo_open_2d`	20/20	20/20	0	0	11.3 ± 0.3	16,415
`demo_four_3d`	20/20	19/20	0	0	12.8 ± 0.6	14,932
`already_at_goals`	20/20	20/20	0	0	0.0 ± 0.0	0
`single_cell_goal`	20/20	20/20	0	0	0.0 ± 0.0	0
`one_reached`	20/20	20/20	0	0	8.3 ± 0.4	14,260
`swap_2d`	20/20	20/20	0	0	5.2 ± 0.3	10,164
`boundary_reverse`	20/20	20/20	0	0	5.3 ± 0.3	11,119
`tiny_2d`	20/20	20/20	0	0	1.0 ± 0.0	3,873
`thin_1d`	20/20	20/20	0	0	7.0 ± 0.0	12,734
`infeasible_altitude`	0/20	0/20	0	0	80.0 ± 0.0	0
`infeasible_wall`	0/20	0/20	0	0	80.0 ± 0.0	0
`infeasible_shared_goal`	0/20	0/20	0	0	80.0 ± 0.0	0
`horizon_one`	0/20	0/20	0	0	1.0 ± 0.0	2,000
`narrow_passage`	18/20	17/20	2	0	21.6 ± 9.4	17,467
`altitude_detour`	20/20	20/20	0	0	9.3 ± 0.3	14,086
`strong_wind`	18/20	19/20	2	0	26.3 ± 8.9	17,260
`three_3d`	19/20	20/20	1	0	14.2 ± 7.3	13,953
`four_2d`	20/20	20/20	0	0	11.7 ± 0.6	15,205

E Supplementary end-to-end cases

These cases fill gaps in the main suites: solvable fixed-altitude single-drone maps (only the infeasible wall case was in the regression suite), fixed-altitude drones on different layers (the setting of the same-layer slip neighbourhood), and a drone whose goal lies inside a teammate's only corridor. Each runs 20 seeds at 2,000 simulations per decision, paired with the search-free base policy (`results/supplement.csv`). The full solver succeeds in 125/140 runs and the base policy in 116/140; there are 0 timeouts, so no run deadlocked, and every failure is a slip collision. The two weaker cases force passages beside obstacles: a one-cell gap in a wall at $\alpha = 0.5$ and a corridor lined with obstacles.

Table 6: Supplementary cases ($n = 20$ each; fixed altitude). No search: the base policy on the same seeds. Obst./Veh./T.o.: obstacle-collision, inter-drone-collision and timeout episodes of the full solver. L: mean ±95% half-width.

Case	Success	Wilson 95% (%)	No search	Obst.	Veh.	T.o.	L
Supplied map, 1 drone	20/20	[84, 100]	20/20	0	0	0	11.2 ± 0.8
Wall with one gap, 1 drone	13/20	[43, 82]	14/20	7	0	0	53.1 ± 11.7
Slalom, 1 drone	19/20	[76, 99]	18/20	1	0	0	28.6 ± 6.7
Five obstacles, 1 drone	20/20	[84, 100]	15/20	0	0	0	16.8 ± 1.8
Two layers, crossing, 2 drones	20/20	[84, 100]	19/20	0	0	0	13.6 ± 1.5
Two layers, 4 drones	20/20	[84, 100]	20/20	0	0	0	12.5 ± 0.9
Goal inside corridor, 2 drones	13/20	[43, 82]	10/20	7	0	0	36.9 ± 15.3

F Algorithm details and model conventions

For an in-grid cell q, let $\mathcal{N}(q)$ be its physical neighbours and $V(q) = |\mathcal{N}(q)|$. For a command towards a free in-grid neighbour,

$$\pi(q) = \frac{d(q)}{d(q)+\alpha}, \quad M(q) = \pi(q) + (V(q)-1)\frac{1-\pi(q)}{B-1},$$
$$\ell(q) = \frac{\pi(q)}{M(q)}, \qquad \sigma(q) = \frac{1-\pi(q)}{(B-1)M(q)}.$$

The obstacle-slip surrogate is $\rho(q) = |\mathcal{N}(q) \cap O|\sigma(q)$. Dijkstra costs use $1 + \lambda\rho(q)$ for entry into a non-goal cell and 1 for entry into the goal, where

$$\lambda = \frac{w(G+|C|)}{\max\{|c_{\text{step}}|,\ 10^{-3}(G+|C|),\ 10^{-9}\}}, \qquad w = 1.$$

With $w = 0$, costs reduce to physical breadth-first distances. At most 150,000 grid cells are mapped; larger grids use a Chebyshev-distance fallback without risk maps. Frozen-goal connectivity checks are cached by drone and reached subset on maps of at most 20,000 cells; larger maps retain the static-distance estimate.

Grid boundaries are not obstacles. With no obstacles, the supplied SciPy distance transform measures distance to a virtual point beyond a grid corner, rather than returning infinity, so slips remain possible. On an obstacle hit, the simulator retains the drone's old cell but returns a terminal flag. The model in B1 maps that outcome to $\perp$; its sampled reward is computed before this mapping. Joint commands are encoded little-endian, $a = \sum_i a_i B^{i-1}$.

At a rollout cutoff after L sampled steps, the normalised return is

$$\sum_{j=0}^{L-1} \gamma^j \frac{r_j}{\kappa} + \gamma^L \frac{\widehat{V}(s_L, h-L)}{\kappa},$$

with zero continuation on terminal outcomes. For a backed-up sample v, Welford's edge update is

$$n \leftarrow n+1, \quad \Delta \leftarrow v - Q, \quad Q \leftarrow Q + \Delta/n, \quad S \leftarrow S + \Delta(v-Q).$$

The pooled root variance is $\hat{\sigma}^2 = \sum_a S_a / \sum_a (n_a - 1)$ over visited actions, or zero if the denominator is zero. This is a sample-based heuristic variance estimate, not a confidence guarantee.

G Factorial condition results

Table 7: All 27 S1 conditions ($n = 20$ each). Continuous entries are mean ±95% interval half-width; Calls counts simulator calls per decision.

n_y	$\|A\|$	$\|O\|$	Success	L	Return	Calls
5	36	0	20/20	13.3 ± 0.9	134.1 ± 3.9	18,413
5	36	2	19/20	21.8 ± 4.6	99.0 ± 23.0	19,010
5	36	5	18/20	23.4 ± 6.1	89.1 ± 30.9	18,539
5	324	0	20/20	12.1 ± 0.6	140.0 ± 2.1	14,951
5	324	2	19/20	15.8 ± 4.9	117.8 ± 35.1	16,166
5	324	5	18/20	18.1 ± 6.7	110.7 ± 34.1	15,664
5	676	0	20/20	12.1 ± 0.7	140.0 ± 2.2	14,730
5	676	2	20/20	13.2 ± 0.9	135.7 ± 3.1	15,330
5	676	5	19/20	16.2 ± 4.9	120.7 ± 24.7	15,284
10	36	0	20/20	13.8 ± 0.7	133.0 ± 2.5	18,368
10	36	2	20/20	17.9 ± 1.2	116.0 ± 4.6	18,599
10	36	5	19/20	28.1 ± 3.9	75.5 ± 28.1	19,998
10	324	0	20/20	12.1 ± 0.6	139.8 ± 2.0	14,939
10	324	2	20/20	13.2 ± 0.7	135.7 ± 2.5	15,738
10	324	5	20/20	15.7 ± 0.8	125.4 ± 3.1	17,113
10	676	0	20/20	12.2 ± 0.6	139.5 ± 2.1	14,759
10	676	2	20/20	13.0 ± 0.7	136.7 ± 2.3	15,264
10	676	5	19/20	16.8 ± 4.8	115.2 ± 33.6	16,457
20	36	0	20/20	13.8 ± 0.7	133.0 ± 2.5	18,368
20	36	2	19/20	20.0 ± 4.6	104.8 ± 23.2	18,891
20	36	5	20/20	25.4 ± 1.4	90.0 ± 3.5	19,787
20	324	0	20/20	12.1 ± 0.6	139.8 ± 2.0	14,939
20	324	2	20/20	13.1 ± 0.7	135.6 ± 2.9	15,738
20	324	5	19/20	17.8 ± 4.7	114.5 ± 24.0	17,184
20	676	0	20/20	12.2 ± 0.6	139.5 ± 2.1	14,759
20	676	2	19/20	15.4 ± 5.0	119.6 ± 35.3	15,672
20	676	5	19/20	16.4 ± 4.9	120.1 ± 24.6	16,513

H Mechanism controls

The 6/18/26-command conditions restrict intended directions only; the physical slip law retains all 26 neighbours. Duplicated labels map to the same physical commands and do not introduce additional slip outcomes.

Table 8: S2 clutter controls ($n_y = 10$, $n = 20$ per row). L: mean ±95% interval half-width; Obst./Veh.: collision episodes.

Commands	α	$\|O\|$	Success	L	Obst.	Veh.
all 26	0.25	0	20/20	12.2 ± 0.6	0	0
		2	20/20	13.0 ± 0.7	0	0
		5	19/20	16.8 ± 4.8	0	1
	0.01	0	20/20	11.0 ± 0.0	0	0
		2	20/20	11.0 ± 0.0	0	0
		5	20/20	11.2 ± 0.2	0	0
axial 6	0.25	0	20/20	13.8 ± 0.7	0	0
		2	20/20	17.9 ± 1.2	0	0
		5	19/20	28.1 ± 3.9	0	1
	0.01	0	20/20	11.6 ± 0.4	0	0
		2	20/20	13.2 ± 0.3	0	0
		5	20/20	14.4 ± 0.5	0	0

Table 9: Duplicated command labels with unchanged physics ($n_y = 10$, $|O| = 2$, $n = 20$). Root: admitted actions; Selected: visit share of the executed root action, which need not be the most-visited action. L: mean ±95% interval half-width.

Labels	$\|A\|$	Success	L	Root	Selected (%)
×1	676	20/20	13.0 ± 0.7	22.6	5.4
×2	2,704	20/20	13.1 ± 0.9	23.0	5.2
×4	10,816	20/20	13.1 ± 0.8	23.0	5.1

I Search-budget results

Table 10: S4 ($n = 20$ per row). L: mean ±95% interval half-width; Root: admitted actions; Depth: mean depth of newly created nodes.

Task (drones)	Sims	Success	L	Root	Depth
Open 2D (1)	16	20/20	11.4 ± 0.4	2.0	2.48
Open 2D (1)	64	20/20	11.4 ± 0.4	4.0	3.07
Open 2D (1)	256	20/20	11.4 ± 0.4	7.7	3.54
Open 2D (1)	1,024	20/20	11.4 ± 0.4	7.7	4.10
Open 2D (1)	2,000	20/20	11.3 ± 0.3	7.7	4.41
Strong wind (1)	16	19/20	21.1 ± 6.8	2.0	1.53
Strong wind (1)	64	17/20	27.9 ± 10.8	4.0	1.72
Strong wind (1)	256	18/20	25.4 ± 9.0	8.0	1.95
Strong wind (1)	1,024	18/20	28.0 ± 8.9	15.0	2.20
Strong wind (1)	2,000	18/20	26.3 ± 8.9	17.3	2.35
Four 3D (4)	16	20/20	13.1 ± 0.7	2.0	1.95
Four 3D (4)	64	19/20	16.2 ± 7.1	4.0	2.29
Four 3D (4)	256	20/20	12.8 ± 0.7	8.0	2.48
Four 3D (4)	1,024	20/20	12.9 ± 0.6	16.0	2.67
Four 3D (4)	2,000	20/20	12.8 ± 0.6	22.8	2.75
Factorial S10/A26/C5 (2)	16	19/20	17.1 ± 4.8	2.0	2.10
Factorial S10/A26/C5 (2)	64	18/20	19.2 ± 6.6	4.0	2.54
Factorial S10/A26/C5 (2)	256	18/20	19.4 ± 6.6	8.0	2.79
Factorial S10/A26/C5 (2)	1,024	18/20	19.1 ± 6.6	15.9	3.03
Factorial S10/A26/C5 (2)	2,000	19/20	16.8 ± 4.8	22.5	3.13

J Generated-layout results

Table 11: S5 marginals: four layouts × ten seeds per level. L: mean ±95% interval half-width; Depth: mean new-node depth. One layout per (N, α) pair limits inference beyond these layouts.

Factor	Level	Success	Wilson 95% CI (%)	L	Depth
Drones N	1	39/40	[87, 100]	10.4 ± 2.7	3.09
	2	38/40	[83, 99]	13.8 ± 3.9	2.72
	3	40/40	[91, 100]	11.7 ± 1.3	2.40
	4	31/40	[62, 88]	21.4 ± 6.8	2.36
Noise α	0.05	40/40	[91, 100]	7.5 ± 0.3	3.45
	0.25	40/40	[91, 100]	8.4 ± 0.5	3.03
	1	36/40	[77, 96]	16.3 ± 4.8	2.23
	2	32/40	[65, 90]	25.0 ± 5.9	1.86

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